

SHUBHAM DESHMUKH

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EDUCATION

Virginia Tech

Master's in Computer Science — GPA: 3.75/4.0

Falls Church, VA

Aug 2023 – May 2025

- Relevant Coursework: Intermediate Data Structures and Algorithms, Software Engineering, Cloud Computing

Vishwakarma Institute of Technology

B.Tech in Electronics and Telecommunications — GPA: 3.56/4.0

Pune, India

Aug 2018 – May 2022

- Relevant Coursework: C Programming, Python, Data Structures & Algorithms, Software Design

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C#, SQL, Bash, Go

Web: React.js, D3.js, Next.js, Node.js, Vue.js, Angular, HTML5, CSS3, Tailwind CSS, REST APIs, GraphQL

Backend: Flask, FastAPI, ASP.NET Core, Spring Boot

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

Cloud: AWS (Lambda, S3, EC2, DynamoDB, CloudFront), GCP, Azure

DevOps: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git, GitLab, Postman, JIRA

Testing: JUnit, PyTest, Selenium

Other: OOP, DSA, System Design, Microservices, Agile (Scrum)

EXPERIENCE

Graduate Research Assistant

Virginia Tech - A3 Lab - Commonwealth Cyber Initiative

Aug 2024 – May 2025

Arlington, VA

- Trained and deployed **ViT & 5 CNN** models using **PyTorch & CUDA**, achieving **$R^2 = 0.89$** for predicting optically active water parameters (IEEE JOE under review).
- Designed a novel **OpenCV** pipeline for **500k+ USGS** images using **Astral & Pixel Filtering**, reducing noise by **40%**.
- Finetuned **U-Net** segmentation model using **CUDA/XLA** and **GMM validation**, reducing processing time by **30%**.

Software Developer Intern

J. Craig Venter Institute

Jun 2024 – Aug 2024

Rockville, MD

- Implemented **D3.js** and **React** to enhance **HSP 3.0 Data Visualization**, improving researchers' engagement by **40%**.
- Optimized **Data pipeline** using **Python, R, Docker, and AWS (Lambda, S3)**, reducing manual intervention by **70%**.
- Enhanced **API queries**, boosting **database efficiency** and significantly decreasing **data processing** time by **20%**.

Software Developer

Wipro Ltd.

May 2022 – Jul 2023

Pune, India

- Led **14 Enterprise Apps** using **ASP.NET Core**, delivering **100% on-time** and enhancing system scalability by **40%**.
- Administered **Tableau Server**, optimizing **analytics, data workflows** and reducing **report generation** time by **15%**.
- Resolved **42 POS** issues, increasing **uptime** by **20%**, improving **transaction speed** and **system reliability**.

Data Science Intern

IFM Engineering Pvt. Ltd.

Aug 2021 – Jan 2022

Pune, India

- Developed a **Real-Time Patient Monitoring System** using **YOLOv4-tiny**, and **Mediapipe** attaining **95% accuracy**.
- Integrated **Deep Learning** models & **Flask Web App**, mitigating operational issues by **30%** and improving efficiency.
- Presented research on a novel **real-time patient monitoring** at **IEEE MysuruCon 2022**, demonstrating **15% faster doctor response** via optimized alert mechanisms.

PROJECTS

Career Compass | [Project Link](#) | *Next.js, FastAPI, AWS EC2, Terraform, LangChain*

Feb 2025 – May 2025

- Engineered a production-ready **LLM RAG system** with **LangChain**, **Groq API (LLaMA-3-70B)**, and **ChromaDB** for interactive career coaching.
- Built **semantic search** pipeline over **O*NET** and job knowledge graphs, improving top-3 retrieval accuracy by **43%**.
- Deployed to **AWS EC2** with **FastAPI** backend, enabling **real-time inference** with **35% latency reduction**.
- Enabled intelligent **agent-like workflows** to adapt dynamically to user inputs with streaming token response.

React Portfolio | [Github](#) | *React, JavaScript, Typescript, HTML, CSS*

Aug 2024 – Sep 2024

- Constructed **Next.js 15 + TypeScript** portfolio with **Tailwind CSS** and **Framer Motion**, boosting visibility **25%**.
- Used **TensorFlow.js**, **React Icons**, and **Heroicons** for dynamic interactive UI elements, improving engagement **30%**.
- Integrated **EmailJS** contact form with smooth animations, deployed **mobile-first design**, increasing accessibility **25%**.